

Solving Problems with Equations

Teacher's Edition

Standard and Targets

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

Boundary Forms are limited to $px \pm q = r$ and $p(x \pm q) = r$. The $p(x \pm q) = r$ form is what this lesson adds, and it is inside the benchmark — parentheses are expected here, not an extension.

Learning targets I can solve two-step equations involving the distributive property. I can solve two-step equations within a real-world context. I can relate the solution of an equation to the context.

Essential questions How do you apply the distributive property to solve two-step equations? How do you relate the solution of an equation to the context it represents?

Building on Lessons 3 and 4, and MA.6.AR.2.2 and MA.6.AR.2.3.

Working toward MA.8.AR.2.1 — multi-step linear equations with rational coefficients, including variables on both sides. Distributing across parentheses is the first move in most of those.

Where this sits in the course Third of five consecutive lessons on MA.7.AR.2.2, and the fifth grade 7 lesson in a row to open Unit 1 — required content these students have not had before. Today adds the $p(x \pm q) = r$ form, which is inside the grade 7 benchmark and is also the first move in most MA.8.AR.2.1 problems. **The distinction on slide 3 — rewriting one side versus operating on both — is the single idea that carries furthest.** Students who leave without it will reverse an inequality sign for the wrong reason in Lesson 9 and lose multi-step equations when they arrive later this year.

Materials Whiteboards and markers. Calculators. Worksheet 7A.01.05. The properties-of-equality anchor chart from Lesson 3.

Lesson at a Glance

Windy Hill periods run **53 minutes**.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up: same and different	2	4 min
Notes I–IV: distributing, a negative outside, two ways to start, when parentheses appear	3–6	12 min
Example 1: $6(x - 2) = 42$, both methods	7–9	5 min
Example 2: the negative outside	10–12	5 min
Example 3: six tickets, each \$1.75 off	13–15	6 min
Whiteboards: 9 questions, each with its reveal	16–33	15 min
IXL — start today's skills	34	5 min

Segment	Slides	Time
Total		53 min

Where the lesson lives: Notes III (slide 5) and Notes I (slide 3). Notes III is the choice of method, which is what the worksheet asks for by name. Notes I's panel is the only place the difference between rewriting a side and operating on both sides is made explicit, and that distinction is what MA.8.AR.2.1 runs on.

Read this before you plan the day. Math Nation's components total 55 minutes before any worksheet. The plan above fits because the Refresher folds into Notes I and the Collaboration becomes Example 2 and the whiteboard sequence. Math Nation's own tip agrees with that first cut: "determine if your students need the support of all or part of the Refresher." For an accelerated class, the distributive property is review.

No partner or group work anywhere in this lesson. Math Nation's version carries six partner instances, second only to Lesson 1's eight. Guided Instruction question 2 is built entirely on the structure: "decide who is Partner A and who is Partner B," each student does one method, then they swap and copy each other's work into the blank column. That is now Example 1, where each student works both methods, which is what a comparison actually requires. The Collaboration is Example 2 and the whiteboard sequence, and both Think-Pair-Shares are whiteboard questions.

Vocabulary

distributive property — $a(b + c) = a \cdot b + a \cdot c$ — the factor outside multiplies every term inside

reciprocal — the fraction turned upside down; multiplying by it undoes dividing

variable — the quantity in a situation that changes

constant — the quantity that stays the same no matter what

A note on the glossary. Math Nation lists no key terms and four additional terms, three of which are property names already on the Lesson 3 anchor chart. Reciprocal is worth a band of its own here, because the efficient route through question 6 of the worksheet is multiplying by $4/3$ rather than dividing by $3/4$.

Slide-by-Slide Teaching Notes

Slide 1. Solving Problems with Equations 1 min

- Post the learning target and call back to Lesson 4's whiteboard question 6 explicitly — $6m + 15$ and $6(m + 15)$ looked almost the same and were different equations.
- Today is why, and what to do about it.

Slide 2. Warm-Up 4 min

- TYPE: Same and different.
- INDEPENDENT.

- ANSWERS: same — the digits, the letter, the order. Different — the left one multiplies BOTH terms by 5 and gives $10y - 35$; the right one multiplies only the $2y$ and gives $10y - 7$. Substitute $y = 1$ into both if the class is stuck: -25 against 3. This replaces Math Nation's opener, which repeated Lesson 4's material.
- OFF THE SLIDE, YOURS TO SAY: Same numbers, same order, same operations. One bracket, and they are not the same expression.

Slide 3. Notes: I. The factor outside multiplies EVERY term inside 3 min

- The panel is the confusion worth heading off: students who have spent two lessons doing the same thing to both sides will try to distribute on both.
- Distributing is a rewrite of one expression, not a move in the balance.
- OFF THE SLIDE, YOURS TO SAY: Distributing rewrites ONE side. It is not something you do to both sides.

Slide 4. Notes: II. A negative outside multiplies both terms too 3 min

- THE NEW DIFFICULTY IN THIS LESSON.
- Write $-4(x - 9)$ on the board and do it slowly, term by term, saying the two signs out loud each time.
- The wrong answer in the panel comes from distributing the 4 and leaving the minus where it was, and it is worth writing up beside the right one.
- OFF THE SLIDE, YOURS TO SAY: $-4x - 36$ is the most common wrong answer in this lesson.

Slide 5. Notes: III. Two ways to start $a(b + c) = d$ 3 min

- Say clearly that this is a choice, not a rule.
- Students who believe there is one correct opening will freeze on the ones where distributing gives ugly numbers.
- Whiteboard questions 4 and 5 are where Method 2 is obviously better.
- OFF THE SLIDE, YOURS TO SAY: If a divides d evenly, Method 2 is usually shorter.

Slide 6. Notes: IV. Parentheses appear when a whole group repeats 3 min

- The panel is the whole of item C in one example, and it is worth reading twice.
- The discount happens to every ticket, so it lives inside the bracket.
- A student who writes $6p - 1.75$ has taken \$1.75 off the whole order once.
- OFF THE SLIDE, YOURS TO SAY: Six tickets, each \$1.75 off: $6(p - 1.75)$, not $6p - 1.75$.

Slide 7. Example 1 1 of the 5 min

- SPOKEN ONLY.
- Method 2 — 6 divides 42 evenly, so dividing first leaves $x - 2 = 7$ and one small step.
- Take both answers; Method 1 is not wrong, it is just longer here.

Slide 8. Example 1 2 of the 5 min

- Work both columns at the board so the equivalence is watched rather than claimed.
- Point at where they meet.
- OFF THE SLIDE, YOURS TO SAY: Same answer. Method 2 took one line fewer because 6 divided 42 evenly.

Slide 9. Example 1 *2 of the 5 min*

- Lesson 3's Notes IV, and it matters more here: a student who distributed wrongly and then checks against their own distributed line will pass their own check.
- OFF THE SLIDE, YOURS TO SAY: Substitute into the bracket form you were given, not into $6x - 12 = 42$.

Slide 10. Example 2 *1 of the 5 min*

- Whiteboards, the distributed line and nothing else.
- Expect $-3x + 4$ from anyone who multiplied only the first term.
- That board is the reason this example exists.

Slide 11. Example 2 *2 of the 5 min*

- Say the two multiplications out loud: -3 times x is $-3x$, and -3 times $+4$ is -12 .
- Then the solving is Lesson 3.
- Watch the last line — dividing 30 by -3 is warm-up territory and students still drop the sign.

Slide 12. Example 2 *2 of the 5 min*

- This check does real work: a student who wrote $-3x + 12$ will get $-3(-10) + 12 = 42$ and see it fail.
- Let that happen rather than correcting the distribution earlier.
- OFF THE SLIDE, YOURS TO SAY: Two negatives in the line — one inside the bracket, one outside. The check is where you find out if you kept track.

Slide 13. Example 3 *2 of the 6 min*

- Whiteboards, equation only.
- The decision is whether the discount is inside the bracket or outside, and it is the whole question.
- Do not accept an answer without the reason.

Slide 14. Example 3 *2 of the 6 min*

- Put both up and read each one back as an English sentence.
- Reading the equation aloud is the check that catches this, and it is faster than any rule about keywords.
- OFF THE SLIDE, YOURS TO SAY: The story says each. So the discount is inside the bracket, and it happens six times.

Slide 15. Example 3 *2 of the 6 min*

- Method 2 again, because 6 divides 46.92 evenly enough.
- The last line is Lesson 4's interpretation step and it is still required — a price, in dollars, and it is the price BEFORE the discount, which is what p was defined to be.

Slide 16. Whiteboards • Question 1 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 17. Whiteboards • Question 1 of 9 — reveal *about 1 minute with the discussion*

- The baseline, and it is Example 1.
- Everything after this changes exactly one thing about it.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 18. Whiteboards • Question 2 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 19. Whiteboards • Question 2 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the sign of the multiplier.
- Dividing 42 by -6 gives -7, not 7, and the answer crosses zero.
- A board saying 9 divided by 6 and ignored the minus.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 20. Whiteboards • Question 3 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 21. Whiteboards • Question 3 of 9 — reveal *about 1 minute with the discussion*

- One feature changed from question 1: the sign INSIDE the bracket.
- Now the last step subtracts instead of adds.
- Questions 2 and 3 move different minus signs and give different answers — worth putting side by side.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 22. Whiteboards • Question 4 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 23. Whiteboards • Question 4 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the multiplier is a decimal, and the total moved to match.
- SAME ANSWER as question 1.
- Method 2 is obviously right here — distributing 0.6 first gives $0.6x - 1.2$ and needless decimals.

- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 24. Whiteboards · Question 5 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 25. Whiteboards · Question 5 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the multiplier is written as a fraction.
- Same answer a third time, because $\frac{3}{5}$ and 0.6 are the same number.
- The point of questions 4 and 5 together is that the FORM of the multiplier changes nothing at all, which is a harder thing to believe than it sounds.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 26. Whiteboards · Question 6 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 27. Whiteboards · Question 6 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT CLOSSES IT.
- One feature changed from question 1: the total, by one.
- Method 2 now gives a fraction immediately, and Method 1 gives $6x = 55$, which is the same fraction one line later.
- Neither method rescues you; the answer simply is not whole.
- Ask which method they would choose NOW — the honest answer is that it no longer matters, which is Notes III's real lesson.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 28. Whiteboards · Question 7 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 29. Whiteboards · Question 7 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the letter is now the SECOND term inside the bracket and it carries a coefficient.
- Divide by 4 and the bracket reads $1 - 3x = 13$, which needs two more moves rather than one.
- THE BOARD TO LOOK FOR SAYS 4: that student reached $-3x = 12$ and read off 4 without the sign, and the check catches it instantly — $4(1 - 12) = -44$, nowhere near 52.

- The other board to look for distributed first and got $4 - 12x = 52$, which is correct and slower; say so, because Notes III's point is that both methods work and one of them is less typing.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 30. Whiteboards · Question 8 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 31. Whiteboards · Question 8 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the total is negative, and the multiplier is a decimal below 1.
- THE BOARD TO LOOK FOR SAYS 1, and the error underneath it is the one worth ten minutes if you have them: dividing by 0.2 makes a number BIGGER, and $-10 \div 0.2$ is -50 , not -2 .
- That student multiplied by 0.2 because dividing by something small feels like it should shrink it.
- Ask how many 0.2s fit into 10 before anyone touches a calculator — fifty — and the answer stops being surprising.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 32. Whiteboards · Question 9 of 9 — ask *about 90 seconds of writing*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 33. Whiteboards · Question 9 of 9 — reveal *about 2 minutes with the closing discussion*

- WRITTEN WORK, and the only question in the round that asks for it.
- With the worksheet gone this is the only place in the period the habit gets built, and IXL takes an answer without ever seeing the work.
- ANSWER: $75 + 66.24h = 339.96$, so $66.24h = 264.96$ and $h = 4$, and $75 + 66.24(4) = 339.96$.
- THE MARK IS FOR THE EQUATION as much as the number — a student who divides 339.96 by 66.24 and gets 5.13 has done arithmetic on the right numbers in the wrong order, and it is the \$75 they forgot to take off first.
- The ugly decimals are deliberate and they are what the calculator is for; the thing a calculator cannot do here is decide what to subtract before dividing.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 34. IXL *5 min*

- IXL.
- The skills are assigned in Google Classroom, not named here, and they are deliberately not ours -- Croix: "The students need questions that aren't exclusively mine."

- Makes them think." Circulate.
- The thing to watch for is a student answering in the box with no paper in front of them: IXL takes an answer and never sees the work, so this is the one part of the period where the habit can quietly lapse.

Worksheet 7A.01.05 — Answer Key

1. $6(x + 4)$	$6x + 24$ 6 multiplies both terms inside.
2. $(b - 9)(-4)$	$-4b + 36$ $(-4)(b) = -4b$ and $(-4)(-9) = +36$. The factor on the right multiplies both terms just the same.
3. $4(1 - 3x) = 52$	$x = -4$ by both methods Method 1: $4 - 12x = 52$, so $-12x = 48$. Method 2: $1 - 3x = 13$, so $-3x = 12$. Check: $4(1 + 12) = 52$.
4. $(3/10)(x + 2) = 12$	$x = 38$ Divide first: $12 \div (3/10) = 40$, so $x + 2 = 40$. Distributing works but puts $3/5$ on the left.
5. $0.2(r - 3) = -10$	$r = -47$ $r - 3 = -50$. Dividing by 0.2 is the same as multiplying by 5.
6. $(3/4)(x + 5\frac{1}{2}) = 5/8$	$x = -4\frac{2}{3}$ $x + 5\frac{1}{2} = (5/8)(4/3) = 5/6$, then $5/6 - 33/6 = -28/6$. Check: $(3/4)(5/6) = 5/8$. The negative answer is correct and students will doubt it.
7. $911.32 = 21.74p + 780.88$	$p = 6$ No parentheses here — a plain two-step equation with large but easy numbers.
8. plumber, equation	$75 + 66.24h = 339.96$ The \$75 is charged once, so it is the constant. The \$66.24 depends on h.
9. plumber, solve	$h = 4$ $66.24h = 264.96$. Check: $75 + 264.96 = 339.96$.
10. plumber, meaning	The plumber worked for 4 hours. Watch for "\$264.96." That is what the hours cost, not the number of hours.

Question 6 is the hardest thing on the page. $(3/4)(x + 5\frac{1}{2}) = 5/8$ gives $x = -4\frac{2}{3}$, and students will assume they are wrong because the answer is negative and the equation had no negatives in it. Have them check rather than restart: $(3/4)(-4\frac{2}{3} + 5\frac{1}{2}) = (3/4)(5/6) = 5/8$.

Question 7 has no parentheses on purpose. After questions 4 through 6, each of which asks the student to choose a method, a plain two-step equation is a useful interruption — it stops students looking for a choice that is not there. The numbers are large but the work is one subtraction and one division.

Question 10 is the one to read tonight. Students who answer \$264.96 have solved the equation and then reported the wrong quantity. That is the same interpret error as Lesson 4's kayak question, and it is the habit this pair of lessons exists to build.

Additional Practice 7A.01.05 — Answer Key

Post this in Google Classroom alongside the practice sheet. It is optional work for students who want more.

1. $4(0.5p + 1.5) = 12$	$p = 3$ Distribute: $2p + 6 = 12$. Or divide by 4: $0.5p + 1.5 = 3$. Check: $4(3) = 12$.
2. $5/2 = (1/4)(z - 8)$	$z = 18$ Multiplying both sides by 4 clears the fraction in one step: $10 = z - 8$.
3. $18 = (3/4)(36.4 - n)$	$n = 12.4$ $18 \times (4/3) = 24$, so $24 = 36.4 - n$. The negative in front of n is the step to watch.
4. $5(2y - 7)$	$10y - 35$
5. $(m + 3.5)(-6)$	$-6m - 21$ Both terms get the negative.
6. cabana total	$\\$882.80$
7. varies with guests	<i>The \$19.40 resort fee, charged once per guest.</i>
8. stays the same	<i>The \$650 cabana rental.</i>
9. how many guests	$650 + 19.40g = 882.80$, so $g = 12$. Twelve guests used the cabana. $19.40g = 232.80$. Check: $650 + 232.80 = 882.80$.
10. Sade's error	<i>She distributed $2/3$ across the left side and also multiplied the right side by $2/3$. The right side should have stayed 9.5.</i> $b = 11.75$ Every line after her second is arithmetically correct, which is what makes this hard to spot. The mistake is in what she was allowed to do, not in what she computed. Check: $(2/3)(14.25) = 9.5$.

Misconceptions to Watch

- **Not recognizing that $p(x + r)$ and $(x + r)p$ are the same.** Notes I item C, worksheet question 2 and Additional Practice question 5. Named first in Math Nation's list. Notes I item C and the whiteboard sequence. (Math Nation TE)
- **Adding or subtracting the coefficient instead of dividing by it.** Also named in their list. (Math Nation TE)
- **Not distributing to both terms inside the parentheses.** Also named, and the single most common error on the worksheet. (Math Nation TE)
- **Not performing the inverse operation on both sides.** Also named in their list. (Math Nation TE)
- **Struggling to relate the solution back to the real-world context.** Also named. Worksheet question 10. (Math Nation TE)
- **Confusing rewriting one side with operating on both sides.** Sade's error. Slide 3 and Additional Practice question 10.
- **Losing the sign on the second term.** $-3(x + 4)$ written as $-3x + 12$. The whiteboard sequence.

- **Taking a per-item discount off only once.** Writing $6p - 1.75 = 46.92$ instead of $6(p - 1.75) = 46.92$. Example 3.

Changes from the Math Nation Lesson

- Every partner and group activity is replaced — six of them. Guided Instruction question 2 was built entirely on partner roles: students were told to decide who was Partner A and who was Partner B, each solved by one method, then swapped and copied the other's work. Each student now works both methods, which is what comparing two methods requires.
- The Additional Practice now asks the question it poses. As printed, the cabana problem states “how many guests used the cabana?” and then questions 7, 8 and 9 ask only for the total bill, the variable cost and the fixed cost. Nobody ever wrote or solved the equation. Question 9 now does: $650 + 19.40g = 882.80$, giving 12 guests.
- Three padded items were merged. Additional Practice questions 2, 4 and 6 were each just “use substitution to verify your solution in question N.” Verification is now part of the solve, which freed room for two distributive-property rewrites and the missing cabana solve.
- The warm-up was replaced. Math Nation's is a decorative green circle pattern that does not survive a photocopier and does not touch the distributive property. Slide 2 is a same-and-different on $5(2y - 7)$ against $5 \cdot 2y - 7$, which is exactly the error this lesson exists to prevent.
- Worksheet question 7 kept its place but gained a substitution check, so it matches the instruction covering the questions around it.
- The Wrap-Up was replaced. Math Nation's Cool Down is “Write a summary of what you learned,” keyed “Answers will vary” — the third lesson in a row with an exit ticket that produces no mathematical evidence. The Ticket Out the Door has students solve with a stated method, distribute a negative, and write an equation that needs parentheses.
- Sade's error was kept and given a rule of its own. Math Nation leaves it as the last question of the Additional Practice with no discussion; here Notes I states the idea it tests — distributing rewrites ONE side; it is not something you do to both — and the question stays on the sheet as the check.
- Answer keys were built for both sheets from the same source file as the student versions, so the two cannot drift apart.
- Nothing in the mathematics was changed.